non-EU companies, provided that they meet the required turnover and presence thresholds.[2632] This assessment aligns with another estimate, which found that U.S. companies could make up 31 percent of an estimated 10,000 U.S., Canadian, and British companies required to begin complying with the CSRD between 2025 and 2029.[2633] However, the number of registrants affected cannot be determined with specificity because the CSRD implementing standards are not fully developed yet, and because the number will depend on factors such as, for example, how many Commission registrants trade on an exchange defined as an EU-regulated market.

Despite uncertainty as to the parameters of other jurisdictions' requirements, the information described above indicates that a meaningful number of Commission registrants may be subject to the climate-related disclosure and reporting requirements of one or more additional jurisdictions. As a result, some registrants subject to the final rules may already have in place, or may be developing, processes and systems to track and disclose aspects of their climate-related risks.

5. Current Market Practices

This section describes current market practices with regard to climate-related disclosure, including disclosures made in Commission filings and in other contexts. This section then describes the use of third-party frameworks in current disclosures; the disclosure of climate-related targets, goals, and transition plans; and the use of third-party assurance.

2632 *See generally* CSRD Q&A, *supra* note 2626; Thibault Meynier, *et al.*, *EU Finalizes ESG Reporting Rules with International Impacts*, Harvard L. Sch. Forum on Corp. Gov. (Jan. 30, 2023), available at https://corpgov.law.harvard.edu/2023/01/30/eu-finalizes-esg-reporting-rules-with-international-impacts/.

2633 Dieter Holger, *At Least 10,000 Foreign Companies to be Hit by EU Sustainability Rules*, Wall St. J. (Apr. 5, 2023), available at https://www.wsj.com/articles/at-least-10-000-foreign-companies-to-be-hit-by-eu-sustainability-rules-307a1406 (retrieved from Factiva database).

We recognize that some aspects of the final rules may overlap with existing disclosure requirements and practices. The incremental costs of the final rules to a specific registrant will depend on the extent to which its disclosures resulting from the final rules overlap with disclosures that would have occurred in the absence of the final rules, as discussed in further detail below.[2634]

a. Climate-Related Disclosures in SEC Filings

The Commission staff reviewed 52,778 annual reports (Forms 10-K and 20-F) submitted from January 1, 2016, until December 31, 2022, to determine how many contain any of the following keywords: "climate change," "climate risk," or "global warming," collectively referred to as "climate-related keywords" throughout this section.[2635] The presence of any of the climate-related keywords in any part of the annual report is indicative of some form of climate-related disclosure.[2636] Table 1 shows the portion of climate-related keywords used in Form 10-Ks and 20-Fs from 2021 through 2022.

2634 *See* section IV.C.3.c, "Factors that Influence Direct Costs." The same point applies similarly to the more general costs imposed by the final rules: those registrants that currently provide (or plan to provide) climate-related disclosures irrespective of the final rules will incur lower incremental costs to the extent that these disclosures overlap with the final rules' requirements.

2635 We follow the approach used in the proposing release except we have excluded 40-F filers because they are not subject to the final rules.

2636 One limitation of using this climate-related keyword search is that it is unable to discern the extent or decision-usefulness of climate-related disclosures, nor can it determine specific sub-topics within climate-related disclosures. For these reasons, the analysis was supplemented by natural language processing ("NLP") analysis, as described later in this section.

Table 1. Filings with Climate-Related Keywords by Form Type

Form	Has Keyword	All Filings	Percent
10-K	4,521	12,846	35%
20-F	662	1,721	38%
Total	5,183	14,567	36%

This table presents the analysis of annual filings submitted to the Commission between Jan. 1, 2021, and Dec. 31, 2022. For each form type, the table indicates how many contain any of the climate-related keywords.

Figure 1 shows that the percentage of Form 10-K and Form 20-F filings with climate-related keywords[2637] has increased between 2016 and 2022. As reflected in Table 1, in more recent filings (i.e., those submitted in calendar years 2021 and 2022) 36 percent of all annual reports contain some climate-related keywords, with a slightly greater proportion (38 percent) among foreign private issuers filing on Form 20-F.[2638] These figures are consistent with data from Bloomberg, which focuses on registrants listed on NYSE and NASDAQ, on ESG reporting. Specifically, using this data, we find that 39 percent of registrants include a discussion of climate related risks in their MD&A section.[2639]

Figure 1 shows that the percentage of Form 10-K and 20-F filings with climate-related keywords[2640] has been increasing between 2016 and 2022. We note that Table 1 reflects the averages of the last two years of the time-series shown in Figure 1.

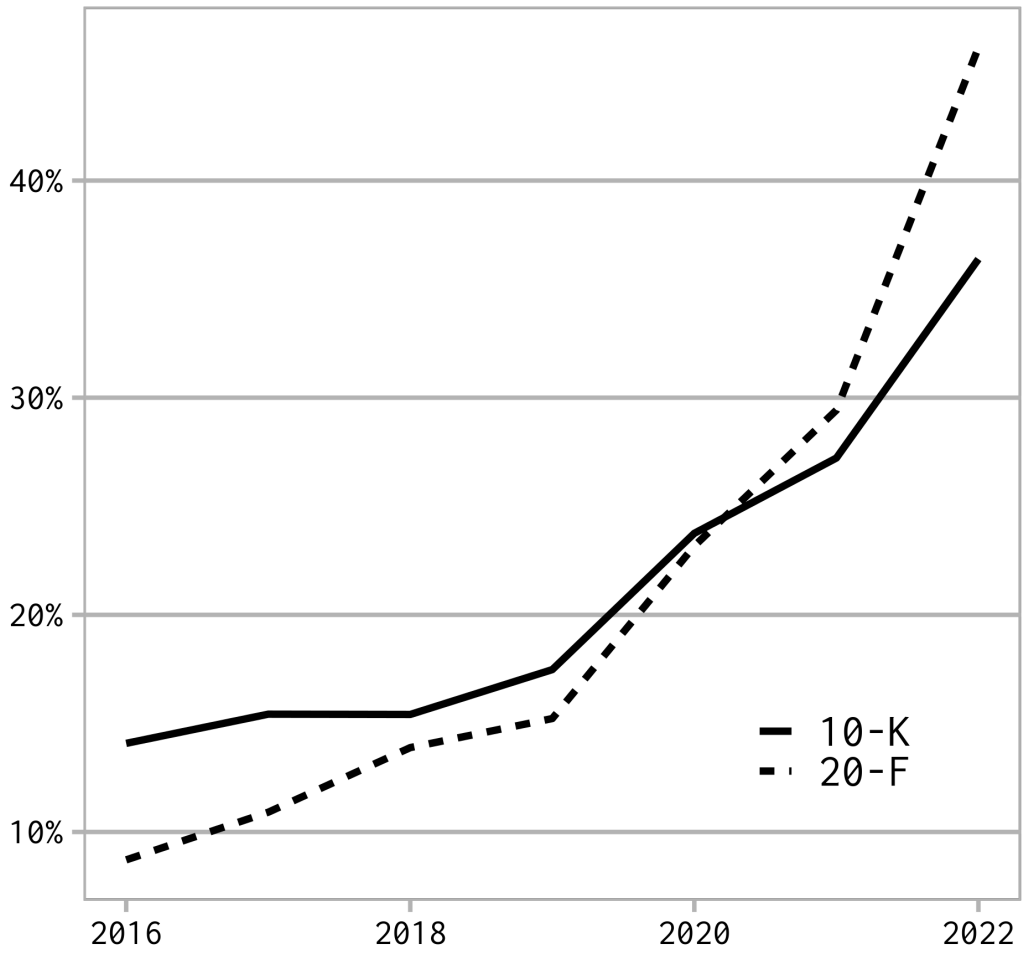
2637 *See supra* note 2636.

2638 Some foreign private issuers may elect to file their annual report on Form 10-K and would thus be classified as "domestic filers" in the following analysis.

2639 Bloomberg reports "[w]hether the Management Discussion and Analysis (MD&A) or its equivalent risk section of registrant's annual report discusses business risks related to climate change." As with other summary statistics presented in this release, these figures may not be representative of all Commission registrants. For example, registrants that are not listed on NYSE or NASDAQ may be less likely to include discussions of climate related risks in their MD&A section.

2640 *See supra* note 2636.

Figure 1. Percentage of Filings with Climate-related Keywords by Form Type



This figure presents the analysis of annual filings on Form 10-K and Form 20-F submitted to the Commission between Jan. 1, 2016, and Dec. 31, 2022. For each form type, the figure plots the percentage of filings containing climate-related keywords.

Table 2 provides a breakdown of more recent filings by accelerated filer status. Among LAFs, 68 percent provided climate-related keywords in 2022, while only 50 percent did so in 2021. Discussions by AFs and NAFs also saw increases over the same period (from 40 to 49 percent and from 16 to 23 percent, respectively).

Table 2. Filings with Climate-related Keywords by Accelerated Filer Status

Year	Filer Status	Has Keyword	All Filings	Percent
2021	LAF	1,063	2,126	50%
	AF	373	936	40%
	NAF	635	3,883	16%
	All	2,071	6,945	30%
2022	LAF	1,726	2,520	68%
	AF	425	863	49%
	NAF	961	4,241	23%
	All	3,112	7,622	41%

This table presents the analysis of annual filings submitted to the Commission between Jan. 1, 2021, and Dec. 31, 2022. For each filer status, the table indicates how many contain any of the climate-related keywords.

Similarly, Table 3 indicates that the inclusion of climate-related keywords by SRCs and EGCs also increased from 2021 to 2022, but that climate change discussions remain less common among these registrants than among registrants that are not SRCs or EGCs.

Table 3. Filings with Climate-related Keywords by SRC/EGC Status

Year	Filer Status	Has Keyword	All Filings	Percent
2021	SRC & EGC	184	2,400	8%
	SRC	744	4,142	18%
	EGC	198	984	20%
	Neither	3,016	6,364	47%
2022	SRC & EGC	440	3,180	14%
	SRC	912	3,724	24%
	EGC	424	1,226	35%
	Neither	4,448	7,114	63%

This table presents the analysis of annual filings submitted to the Commission between Jan. 1, 2021, and Dec. 31, 2022. Filer status SRC, EGC, small emerging growth companies ("SRC & EGC"), and large non-EGC and non-SRC companies ("Neither"). For each filer status, the table indicates how many contain any of the climate-related keywords.

Table 4 (presented as a graph in Figure 2) provides a breakdown of the recent filings by industry and shows that the industries with the highest percentage of annual reports containing climate-related disclosure include electric services, maritime transportation, steel manufacturing, paper and forest products, and oil and gas, among others.

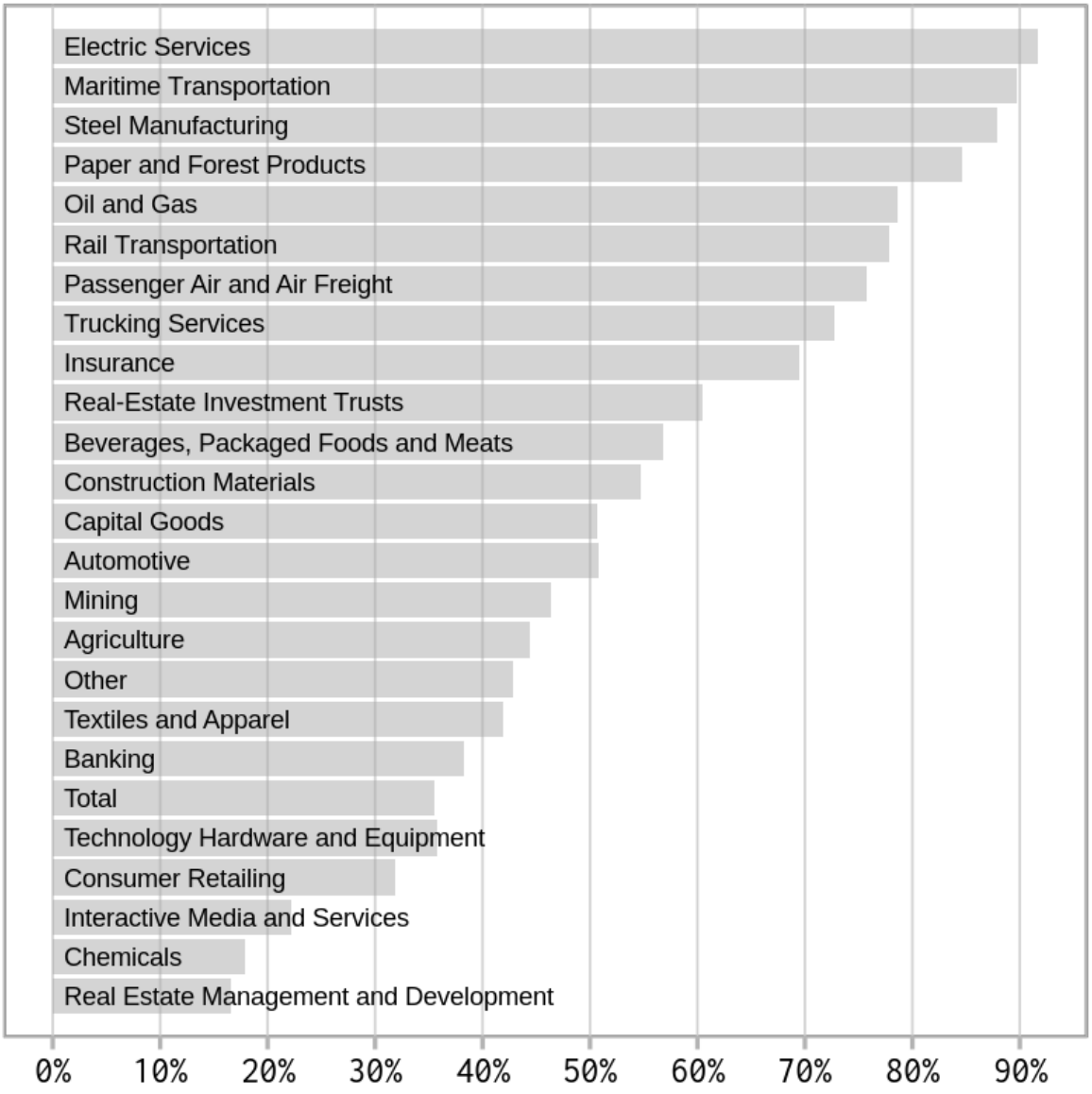
Table 4. Filings with Climate-related Keywords by Industry

Industry	Has Keyword	All Filings	Percent
Electric Services	144	157	92%
Maritime Transportation	114	127	90%
Steel Manufacturing	29	33	88%
Paper and Forest Products	44	52	85%
Oil and Gas	350	445	79%
Rail Transportation	14	18	78%
Passenger Air and Air Freight	50	66	76%
Trucking Services	32	44	73%
Insurance	189	272	69%
Real-Estate Investment Trusts	292	483	60%
Beverages, Packaged Foods and Meats	138	243	57%
Construction Materials	128	234	55%
Automotive	34	67	51%
Capital Goods	123	243	51%
Mining	154	332	46%
Agriculture	32	72	44%
Other	622	1,454	43%
Textiles and Apparel	31	74	42%
Banking	558	1,460	38%
Technology Hardware and Equipment	618	1,725	36%
Consumer Retailing	392	1,229	32%
Total	5,188	14,593	36%

This table presents the analysis of annual filings submitted to the Commission between Jan. 1, 2021, and Dec. 31, 2022. For each industry, the table indicates how many contain any of the climate-related keywords.

Figure 2 provides a breakdown by industry of use of climate-related keywords.

Figure 2. Percentage of Filings with Climate-related Keywords by Industry



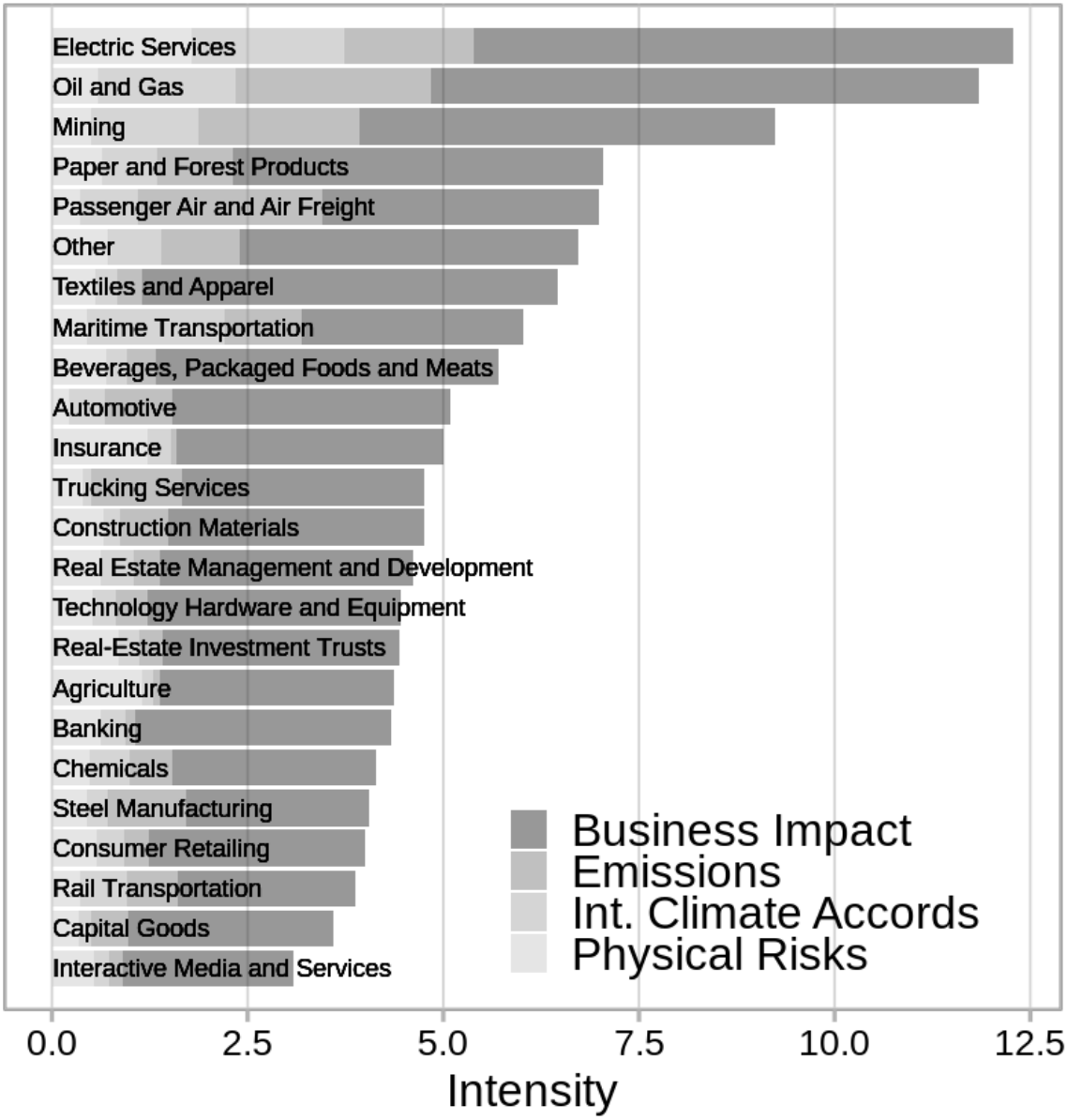
Using the same sample of recent annual reports, Commission staff conducted additional analysis using NLP, which can provide insight on the semantic meaning of individual sentences

within registrants' climate-related disclosures and classify them into topics (i.e., clusters).[2641] The NLP analysis suggests that climate-related disclosures can be broadly organized into four topics: business impact, emissions, international climate accords, and physical risks. The analysis finds significant variation, both within the quantity and content, of climate-related disclosures across industries, as shown in Figures 3 and 4.

Figure 3 presents the intensity of disclosure for domestic annual report filings (Form 10-K). The intensity refers to sentences per registrant, which is calculated by taking the aggregate number of sentences in an industry and dividing it by the total number of registrants within the industry (including those that do not include any climate-related keywords). Thus, the intensity represents a more comparable estimate across industries. Figure 3 shows that registrants in the following industries have the highest intensity of disclosures: oil and gas, electric services, and mining. The majority of these disclosures addressed business impact, followed by emissions, international climate accords, and physical risks. Figure 4 presents the corresponding information for foreign annual report filings (Form 20-F). The foreign filings contain considerably higher intensity of climate-related keywords. For example, Form 10-K filers in the oil and gas industry have approximately 12 sentences per filing containing climate-related keywords while foreign filers in the same industry devote approximately 75 sentences per filing containing climate-related keywords. Overall, the analysis indicates that the majority of the disclosure for both domestic and foreign filings is focused on transition risks, with comparatively fewer mentions of physical risk.

[2641] The specific NLP method used in this analysis is word embedding, which utilizes Google's publicly available, pre-trained word vectors that are then applied to the text of climate-related disclosures within regulatory filings. While this NLP analysis can be used to identify the general topic and the extent of disclosures, it is limited in its ability to discern the decision-usefulness of disclosures from investors' perspective.

Figure 3. Clustered Intensity by Industry for Domestic Filings (Form 10-K)



This figure presents the analysis of Form 10-K annual filings submitted to the Commission between Jan. 1, 2021, and Dec. 31, 2022. NLP was used to analyze sentences contained within the annual filings and classify them into four broad topics (i.e., clusters): business impact, emissions, international climate accords, and physical risks. Intensity refers to the average number of sentences per registrant, which is calculated by taking the aggregate number of relevant sentences in an industry and dividing it by the total number of registrants within the industry.